



2025 FRM Part II

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风险管理与投资管理

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4. Risk Management and Investment Management

4.1. Key Point: Factor Theory

4.1.1. 重要知识点

4.1.1.1. Factors and CAPM

- **Factors definition**
 - Factors are to assets what nutrients are to food.
 - Factor risks are driving force behind assets' risk premiums.
 - Factors matter, not assets.
 - Assets are bundles of factors.
 - Different investors need different risk factors.
- **Factor risks are bad.**
- **Factor theory specifies different types of underlying factor risk.**
- **CAPM and its assumptions**
 - Assumptions:
 - ✓ No transaction costs.
 - ✓ Assets are infinitely divisible.
 - ✓ The absence of personal income tax.
 - ✓ An individual can not affect the price of a stock by his trading.
 - ✓ Investors' decision making solely depend on terms of returns and standard deviations of the returns.
 - ✓ Unlimited short sales are allowed.
 - ✓ Unlimited lending and borrowing at the riskless rate.
 - ✓ All investors have identical expectations.
 - ✓ All assets are marketable.
 - CAPM risk premiums depend only on the assets' beta, which means only one factor – market portfolio.
 - Shortcomings of CAPM come from its assumptions.
 - Basic intuitions of the CAPM still holds true: risk premiums are compensation for investors' losses during bad times.
- **Efficient Market Theory and how markets can be inefficient. Losses during bad times are compensated for by high returns.**

4.1.2. 基础题

Q-1. Which of the following concepts most likely to be a factor?

- A. US Treasury Bill.
- B. Corporate Bonds.
- C. Private Equity.

D. Hedge Fund.

Q-2. Andrew Ang develops an analogy, writing "factors are to assets what nutrients are to food." His theory of factor risk premiums includes each of the following three ideas EXCEPT which is not in the theory?

- A. Assets are bundles of factors (just as most foods are combinations of nutrients)
- B. Factors do matter but asset classes do not (just as healthy eating is about the nutrients not the labels)
- C. Different investors prefer and/or need different factors (just as different people have different nutritional needs)
- D. Because factors represent different good times, most investors should seek exposure to most investable factors (just as most people should seek a balanced diet of most nutrients)

Q-3. In regard to the capital asset pricing model (CAPM), which of the following assumptions (or implications) of the CAPM is a genuine success such that it is both true in practice and useful to us?

- A. Information is costless and available to all investors: technology has reduced information friction to roughly zero
- B. Risk is factor exposure: The risk of an individual asset is measured in terms of the factor exposure of that asset
- C. Investors have mean-variance utility: asset owners care only about means (which they like) and variances (which they dislike)
- D. Investors have homogeneous expectations: investors have identical expectations with respect to the necessary inputs into the portfolio decision

Q-4. Which of the following statements is a limitation of the capital asset pricing model (CAPM)?

- A. Investors have a single period investment horizon.
- B. The market is not transparent
- C. Investors have heterogeneous expectations.
- D. People need to pay a liquidity premium to do transaction.

Q-5. Assets that have big profits during periods of low market returns have:

- A. Low betas and low risk premiums.
- B. Low betas and high risk premiums.
- C. High betas and low risk premiums.

D. High betas and high risk premiums.

Q-6. Which behavior does asset payoffs and “bad times” events would most likely perform?

- A. The expected payoff of an asset in bad times is unrelated to the asset’s expected return, because arbitrageurs eliminate any expected return potential.
- B. The expected payoff of an asset in bad times is unrelated to the asset’s expected return, because it depends on investor preferences.
- C. The higher the expected payoff of an asset in bad times, the higher the asset’s expected return.
- D. The higher the expected payoff of an asset in bad times, the lower the asset’s expected return

4.2. Key Point: Factors

4.2.1. 重要知识点

4.2.1.1. Factors

➤ Factors Investing

- Value Investing and Value Premium
- A value-growth strategy is long value growth stocks and short growth stocks.
- A value stock has a high book-to-market ratio, a growth stock has a low book-to-market ratio.

➤ Macroeconomic risk factors

- Economic growth, inflation, and volatility are the three most important macro factors that affect asset prices.
- Rather than level of a factor, it is the unanticipated change to a risk factor that affects asset prices.

➤ Mitigating volatility risk and Challenges

- Two basic approaches to mitigate volatility risk:
 - ✓ Invest in less-volatile assets like bonds.
 - ✓ Buy volatility protection in the derivatives market.

➤ Dynamic risk factor

- The Fama-French model explains asset returns on three dynamic factors:
 - ✓ Traditional CAPM market risk factor.
 - ✓ A factor that captures size effect (SMB or small cap minus big cap)
 - ✓ A factor that captures value/growth effect (HML or high book-to-market value minus low book-to-market value).

➤ Value and momentum investment strategies

- A momentum strategy is long “winners” and short “losers”.
- Value strategy is a negative feedback strategy, momentum is a positive feedback strategy.
- Momentum strategies are subject to crashes.
- **Different strategy based on different situations and needs.**

4.2.2. 基础题

Q-7. A high book-to-market value ratio is indicative of a:

- A. Small-cap stock.
- B. Large-cap stock.
- C. Value stock.
- D. Growth stock

Q-8. Which of the following statements is TRUE about the momentum factor?

- A. Momentum is a negative feedback strategy which is inherently stabilizing
- B. The momentum factor is observed in equities but is NOT observed in bonds, commodities and real estate
- C. Momentum investing by definition is an anti-value strategy; correlations between HML and WML are strongly negative
- D. The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on either size or value

Q-9. Which of the following is a factor in the Fama-French three-factor model?

- A. Investment Growth.
- B. The small capitalization minus big capitalization risk factor.
- C. The winners minus losers risk factor.
- D. Inflation.

Q-10. Which of the following investment strategies destabilizes asset prices most?

- A. A value strategy.
- B. A size investment strategy.
- C. A momentum investment strategy.
- D. Value, momentum, and size strategies all stabilize asset prices.

4.3. Key Point: Alpha and the Low-Risk Anomaly

4.3.1. 重要知识点

4.3.1.1. Alpha and the Low-Risk Anomaly

- Alpha, tracking error, information ratio, Sharpe ratio.
 - Alpha is the average performance of an investor in excess of their benchmark.
 - The standard deviation of excess return is known as tracking error.
 - Information ratio is the ratio of alpha to its tracking error.

$$IR = \frac{\alpha}{\sigma_{\alpha}}$$
 - If an investor is using risk free rate as benchmark, Sharpe ratio is:

$$SR = \frac{E(R_P) - R_f}{\sigma_P}$$
- Ideal Benchmark should be:
 - Well defined.
 - Tradable.
 - Replicable
 - Adjusted for risk.
- Grinold's fundamental law of active management suggests a tradeoff between the number of investment bets placed(breadth) and the required degree of forecasting accuracy(information coefficient)

$$IR \approx IC \times \sqrt{BR}$$
- Factor Regression
 - Traditional CAPM, APT, Fama-French.
- Application of Factor Regression:
 - Style analysis
- Alphas for nonlinear strategies
- Risk Anomaly
 - Risk anomaly – the stocks with low betas and low volatilities have high returns.
 - Explanations:
 - ✓ Data mining
 - ✓ Leverage constraints
 - ✓ Agency Problems
 - ✓ Preferences
- A comprehensive explanation for risk anomaly is elusive.

4.3.2. 基础题

Q-11. Which of the following statements is incorrect concerning the low-risk anomaly?

- A. The low-risk anomaly conflicts with the CAPM.
- B. The firms with higher beta perform indifferently with the lower beta firms.
- C. The low-risk anomaly point to a negative relationship between risk and reward.
- D. The low-risk anomaly suggests that low-beta stocks will outperform high-beta stocks.

Q-12. Which of the following statements is not true regarding benchmark?

- A. A benchmark should be well-defined.
- B. A benchmark should be replicable.
- C. A benchmark should be equally applied to all risky assets irrespective of their risk exposure.
- D. A benchmark should be tradeable.

Q-13. Following Grinold's fundamental law of active management, one should find:

- A. Sector allocation is the most important factor in active management.
- B. To maximize the information ratio, active investors need to either have high-quality predictions or place a large number of investment bets in a given year.
- C. A small number of investment bets decreases the chances of making a mistake and, therefore, increases the expected investment performance.
- D. Investors should focus on increasing only their predictive ability relative to stock price movements.

Q-14. Why would an investor include multiple factors in a regression study?

- I. To attempt to improve the adjusted R^2 measure.
 - II. To search for a benchmark that is more representative of a portfolio's investment style.
 - III. To increase the tests of statistical significance.
- A. I only.
 - B. Both I and III.
 - C. Both I and II.
 - D. I, II, and III.

Q-15. Regarding of the risk anomaly, which of the following characteristics is a possible reason?

- A. Investor preferences.
- B. The presence of highly leveraged retail investors.
- C. Lack of short selling constraints for institutional investments.
- D. Lack of tracking error constraints for institutional investments.

Q-16. Grinold's fundamental law says that the expected information ratio of an active investment strategy, IR, is given by the information coefficient (IC) multiplied by the square root of the breadth, BR, which is a measure of diversification. In words, as Grinold says, "High information ratios follow from high levels of skill and/or high breadth or diversification." The presumption is that information ratios are the primary measure of value added. In a subsequent development the transfer coefficient, TC, was incorporated as a multiplier such that the extended fundamental law is given by $IR = IC \times \sqrt{BR} \times TC$. The transfer coefficient, TC, is a measure of efficiency: to what extent do transaction costs and/or other constraints cause the actual portfolio to vary from the optimal portfolio?

Which of the following is the BEST measure of breadth, BR, in the fundamental law?

- A. The number of bets per year
- B. The ratio of active return to active risk
- C. The number of positions (aka, holdings) in the portfolio
- D. The number of securities in the relevant global index, e.g., S&P 1500, Russel 3000, or Wilshire 5000

4.4. Key Point: Portfolio Construction

4.4.1. 重要知识点

4.4.1.1. Portfolio Construction

- Portfolio Construction Techniques
 - Screens
 - Stratification
 - Linear Programming
 - Quadratic Programming

4.4.2. 基础题

Q-17. Which statement about risk control in portfolio construction is correct?

- A. Quadratic programming allows for risk control through parameter estimation but manager is concerned about a potential increase in investor redemptions and wants to assess the effects of such an event on the fund's liquidity. The manager asks a junior analyst to estimate the average number of days required to liquidate certain generally requires many more inputs estimated from market data than other methods require.
- B. The screening technique provides superior risk control by concentrating stocks in selected sectors based on expected alpha.
- C. When using the stratification technique, risk control is implemented by overweighting

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the categories with lower risks and underweighting the categories with higher risk.

- D. When using the linear programming technique, risk is controlled by selecting the portfolio with the lowest level of active risk.

Q-18. A portfolio manager is constructing an equity portfolio that will include several equities from multiple industries. The manager aims to achieve mean/variance optimization and asks a group of analysts to perform alpha analysis and also to estimate other portfolio inputs. As part of the alpha analysis, the analysts adjust alphas based on the portfolio's constraints and exclude very large positive and negative alphas from their calculations. Additionally, the analysts plan to remove biases or undesirable bets from the alphas. Which of the following procedures should the analysts apply to execute this plan?

- A. Stratification
- B. Neutralization
- C. Scaling
- D. Trimming

Q-19. The risk committee of an equity mutual fund is reviewing a portfolio construction technique proposed by a new portfolio manager who has recently been allocated capital to manage. The fund typically grants its portfolio managers flexibility in selecting and implementing appropriate portfolio construction procedures but requires that any methodology adopted fulfill key risk control objectives set by the firm. Which of the following is correct regarding a portfolio construction technique and its approach to risk control?

- A. The quadratic programming technique takes into account additional risk parameters compared to other major portfolio construction techniques but also requires more inputs, which leads to more noise.
- B. The screening technique ranks stocks by risk-adjusted alpha but it does not apply any additional risk control measures such as weighting the selected stocks by their relative capitalization.
- C. The stratification technique splits the list of stocks into categories and maintains risk control by overweighting the categories with lower risks and underweighting the categories with higher risks.
- D. The linear programming technique focuses on the pair-wise correlations of stocks rather than characterizing each stock along multiple dimensions of risk such as size, industry, volatility, or beta.

Q-20. John is a portfolio manager managing a fund that aims to outperform a benchmark index.

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He uses the concept of the optimal risk aversion coefficient to balance the trade-off between risk and return. Over the past year, John's portfolio achieved an Information Ratio (IR) of 0.8, and the Tracking Error (TE) relative to the benchmark was 12%. What is the optimal risk aversion coefficient for John's portfolio?

- A. 0.033
- B. 0.067
- C. 0.133
- D. 0.267

4.5. Key Point: Portfolio Risk Measures

4.5.1. 重要知识点

4.5.1.1. Portfolio Risk Measures

➤ Portfolio VaR

- VaR for uncorrelated positions ($\rho=0$):

$$\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$$

- Undiversified VaR ($\rho=1$)

$$\text{VaR}_p = \text{VaR}_1 + \text{VaR}_2$$

➤ Marginal VaR, Incremental VaR and Component VaR

- $\text{MVaR}_A = \frac{\partial \text{VaR}_p}{\partial V_A}$

$$= z_\alpha \times \frac{\text{Cov}(R_A, R_p)}{\sigma_p}$$

$$= z_\alpha \times \rho_{A,p} \times \sigma_A$$

$$= z_\alpha \times \beta_{A,p} \times \sigma_p$$

$$= \frac{\text{VaR}_p}{V_p} \times \beta_{A,p}$$

- Incremental $\text{VaR}_A \approx \text{MVaR}_A \times W_A$ (any amount)

$$\text{Component VaR}_A = \text{MVaR}_A \times V_A$$

- Global Minimum Portfolio: $\text{MVaR}_i = \text{MVaR}_j$

- Optimal Portfolio:

$$\frac{\text{Position i return} - \text{risk free rate}}{\text{MVaR}_i} = \frac{\text{Position j return} - \text{risk free rate}}{\text{MVaR}_j}$$

- Liquidity Duration: It is an approximation of the number of days necessary to dispose of a portfolio's holdings without a significant market impact.

$$\text{LD} = \frac{\text{number of shares of a security}}{\text{desired max daily volume}(\%) \times \text{daily volume}}$$

4.5.2. 基础题

Q-21. A wealth management firm has a portfolio consisting of USD 48 million invested in US equities and USD 35 million invested in emerging markets equities. The 1-day 95% VaR for each individual position is USD 1.2 million. The correlation between the returns of the U.S. equities and emerging markets equities is 0.36. While rebalancing the portfolio, the manager in charge decides to sell USD 8 million of the US equities to buy USD 8 million of the emerging markets equities. At the same time, the CRO of the firm advises the portfolio manager to change the risk measure from 1-day 95% VaR to 10-day 99% VaR. Assuming that returns are normally distributed and that the rebalancing does not affect the volatility of the individual equity positions, by how much will the portfolio VaR increase due to the combined effect of portfolio rebalancing and change in risk measure?

- A. USD 4.529 million
- B. USD 6.258 million
- C. USD 7.144 million
- D. USD 7.223 million

Q-22. The bank's trading book consists of the following two assets:

Asset	Annual Return	Volatility of Annual Return	Value
A	10%	25%	100
B	20%	20%	50

Correlation (A, B) = 0.2

How would the daily VaR at 99% level change if the bank sells 50 worth of asset A and buys 50 worth of asset B?

Assume there are 250 trading days in a year. ($\mu_{1\text{-day}} = 0$)

- A. 0.2286
- B. 0.4578
- C. 0.7705
- D. 0.7798

Q-23. A portfolio manager is evaluating the risk profile for a portfolio of stocks. Currently, the portfolio is valued at CAD 20 million and contains CAD 5 million in stock XYZ. The standard deviation of returns of stock XYZ is 15% annually and that of the overall portfolio is 12% annually. The correlation of returns between stock XYZ and the portfolio is 0.3. Assuming the portfolio manager uses a 1-year 99% VaR and that returns are normally distributed, what is the estimated component VaR of stock XYZ?

- A. CAD 162,972
- B. CAD 234,906
- C. CAD 523,350

D. CAD 632,152

Q-24. Consider a USD 1 million portfolio with an equal investment in two funds, Alpha and Omega, with the following annual return distributions:

Fund	Expected Return	Volatility
Alpha	5%	20%
Omega	7%	25%

Assuming the returns follow the normal distribution and that there are 252 trading days per year, what is the maximum possible daily 95% Value-at-Risk (VaR) estimate for the portfolio? ($\mu_{1\text{-day}} = 0$)

- A. USD 16,587
- B. USD 23,316
- C. USD 23,459
- D. USD 32,973

Q-25. A portfolio consists of two positions. The VaR of the two positions are \$10 million and \$20million. If the returns of the two positions are not correlated. The VaR of the portfolio would be closest to:

- A. \$5.48million
- B. \$15.00million
- C. \$22.36million
- D. \$25.00million

Q-26. A portfolio is composed of two securities and has the following characteristics:

Investment in X: USD 1.8 million

Investment in Y: USD 3.2 million

Volatility of X: 8%

Volatility of Y: 4%

Correlation between X and Y: 15%

The portfolio diversified VaR at the 95% confidence level is closest to:

- A. \$14,074
- B. \$206,500
- C. \$404,740
- D. \$339,722

Q-27. The manager of the Beta Balance fund, a balanced global equity and fixed-income

portfolio, believes that globalization is causing the correlations of equity and fixed-income returns across different markets to rise over time. He decides to adjust the correlations in his VaR analysis for the coming year to reflect the higher correlations he expects. If his expectation turns out to be incorrect, what is the most likely result?

- A. There will be no impact on the portfolio because VaR is only a prediction, and portfolio return depends on what actually happens.
- B. The portfolio return will be lower than it should have been, given the expected risk level, because asset allocation will not have been optimal.
- C. The risk of the portfolio will have been understated because of the incorrect estimate of correlation among global markets.
- D. The portfolio return will be higher than it should have been, given the expected risk level, because of the higher correlation among asset classes

The next two questions are based on the following information.

A risk manager assumes that the joint distribution of returns is multivariate normal and calculates the following risk measures for a 2-asset portfolio:

Asset	Position	Individual VaR	Marginal VaR	VaR Contribution
1	USD 100	USD 23.3	0.176	USD 17.6
2	USD 100	USD 46.6	0.440	USD 44.0
Portfolio	USD 200	USD 61.6		USD 61.6

Q-28. If asset 1 is dropped from the portfolio, what will be the reduction in portfolio VaR?

- A. USD 15.0
- B. USD 38.3
- C. USD 44.0
- D. USD 46.6

Q-29. Let $\beta_i = \rho\sigma_i/\sigma_p$, where ρ denotes the correlation between the return of asset i and the return of the portfolio, σ_i is the volatility of the return of asset i and σ_p is the volatility of the return of the portfolio. What is β_2 ?

- A. 0.714
- B. 1.429
- C. 1.513
- D. Cannot determine from information provided.

Q-30. Consider the following two asset portfolios:

Asset	Position Value	Return Standard Deviation (%)	Beta
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	(In Thousands of USD)		
A	400	3.60	0.5
B	600	8.63	1.2
Portfolio	1,000	5.92	1

Calculate the component VaR of asset A and marginal VaR of asset B, respectively, at the 95% confidence level.

- A. USD 21,773 and 0.1306
- B. USD 21,773 and 0.1169
- C. USD 19,477 and 0.1169
- D. USD 19,477 and 0.1306

Q-31. A risk analyst is evaluating the risks of a portfolio of stocks. Currently, the portfolio is valued at EUR 200 million and contains EUR 15 million in stock A. The standard deviation of returns of stock A is 16% annually and that of the overall portfolio is 21% annually. The correlation of returns between stock A and the portfolio is 0.37. Assuming the risk analyst uses a 1-year 99% VaR and that returns are normally distributed, how much is the component VaR of stock A?

- A. EUR 2.066 million
- B. EUR 2.326 million
- C. EUR 5.582 million
- D. EUR 7.327 million

Q-32. A portfolio manager currently holds 8,000 shares of GF Inc. in a particular portfolio. The daily volume of GF shares traded on the stock exchange is 2,000 shares. Additionally, on any given day, the portfolio manager wishes to trade no more than 25% of the daily trading volume of GF. Which of the following amounts is closest to the liquidity duration of GF in this portfolio?

- A. 0.06
- B. 0.40
- C. 6.50
- D. 16.00

Q-33. A portfolio manager wants to invest a small amount of new money that has recently come into a fund. The fund is benchmarked to an index and, rather than adding a new holding, the manager is considering increasing the holdings of one of the four assets described in the following table:

Asset	Expected Return	Beta to the Index	Beta to the Portfolio
A	12%	1.2	0.90
B	10%	0.7	0.90
C	10%	0.6	0.85
D	8%	0.3	1.10

The portfolio manager wants to select the asset that has the lowest marginal VaR as long as its Treynor ratio is at least 0.1. Assuming the risk free rate is 2%, which asset should the portfolio manager select?

- A. Asset A
- B. Asset B
- C. Asset C
- D. Asset D

Q-34. A hedge fund manager is concerned about a potential increase in investor redemptions and wants to assess the effects of such an event on the fund's liquidity. The manager asks a junior analyst to estimate the average number of days required to liquidate certain securities in the fund. The analyst uses the information presented below to make this estimation for four securities:

Security	Market value of security in the fund (CNY million)	Shares of security in the fund	Average daily trading volume of security	Maximum daily volume allowed for liquidation (expressed as a percentage of average daily trading volume)
A	93.00	500,000	522,000	22%
B	173.04	420,000	1,328,000	12%
C	58.88	256,000	710,000	18%
D	28.80	640,000	848,000	20%

Which of the four securities listed above is expected to take the longest to liquidate?

- A. Security A
- B. Security B
- C. Security C
- D. Security D

Q-35. A portfolio manager holds a USD 4 million equity portfolio that consists of two equities, TOM and JRY, whose returns are uncorrelated. Additional information on TOM and JRY is given below:

	Current weight	Expected return (%)	Volatility of returns	Marginal VaR
TOM	40	14.0	0.08	0.0621
JRY	60	17.5	0.10	0.1456

The manager would like to construct the optimal portfolio that maximizes the portfolio's Sharpe

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ratio and considers the following portfolio weighting schemes:

Weighting scheme	Proposed weight (%)		Marginal VaR	
	TOM	JRY	TOM	JRY
1	50	50	0.0833	0.1042
2	56	44	0.0941	0.0959
3	75	25	0.1218	0.0508

Which weighting scheme is the closest to providing an optimal portfolio?

- A. Current weighting scheme
- B. Weighting scheme 1
- C. Weighting scheme 2
- D. Weighting scheme 3

Q-36. XYZ Investment Firm is interviewing four portfolio managers for a senior portfolio management position. The firm has two assets, Asset A and Asset B, with the following characteristics:

Asset A: Expected return = 8%, Volatility = 15%

Asset B: Expected return = 6%, Volatility = 10%

The risk-free rate is 2%.

The goal is to determine the optimal allocation between Asset A and Asset B that maximizes the Sharpe Ratio of the portfolio. After the interview, each of the four candidates submitted their proposed allocation ratios of the portfolio and other relevant data to support their decisions. The results are as follows:

Candidate	Asset A's weight	Asset B's weight	MVaR _A	MVaR _B
1	55%	45%	0.0546	0.0546
2	45%	55%	0.0584	0.0490
3	60%	40%	0.0660	0.0440
4	40%	60%	0.0742	0.0581

Based on the above results, which candidate should XYZ Investment Firm hire as the portfolio manager?

- A. Candidate 2
- B. Candidate 3
- C. Candidate 4
- D. Candidate 1

4.6. Key Point: Portfolio Risk Management

4.6.1. 重要知识点

4.6.1.1. tracking error

- Relevant return is the tracking error (TE), which is excess return of asset over benchmark.

$$TE = R_P - R_B$$

$$TEV = \sigma(e) = \sqrt{\sigma_P^2 - 2\rho\sigma_P\sigma_B + \sigma_B^2}$$

4.6.1.2. policy mixed VaR

- the policy mix risk is the risk of a dollar loss owing to the policy mix selected by the fund.

4.6.1.3. Surplus at Risk

- Surplus (S) is the difference between the value of the assets (A) and the liabilities (L). The change in the surplus (ΔS) is equal to the change in assets (ΔA) minus the change in liabilities (ΔL). If we normalize by the assets, the return on the surplus is given by:

$$\text{Expected surplus} = A \times (1 + R_A) - L \times (1 + R_L)$$

$$\sigma_{\text{Surplus}} = \sqrt{A^2\sigma_A^2 + L^2\sigma_L^2 - 2A\sigma_AL\sigma_L\rho}$$

$$\text{Surplus at risk} = z_\alpha \times \sigma_{\text{Surplus}}$$

- Funding risk should be measured as the potential shortfall in surplus over the horizon, this is sometimes called surplus at risk.

4.6.1.4. Risk Budgeting

- **Budget Risk across Asset Classes:** Budgeting risk across asset classes means selecting assets whose combined VaRs are less than the total allowed. The budgeting process would examine the contribution each position makes to the portfolio VaR.
- **Budget Risk across Active Managers:** For allocating across active managers, if the tracking errors of the managers are independent of each other, it can be shown that the optimal allocation is achieved with the following formula:

$$\text{weight of portfolio managed by manager } i = \frac{IR_i \times (\text{Portfolio's tracking error volatility})}{IR_P \times (\text{Manager's tracking error volatility})}$$

- For a given group of active managers, the weights may not sum to one. The remainder of the weight can be allocated to the benchmark, which has no tracking error.

4.6.2. 基础题

- Q-37.** You are evaluating the performance of Valance, an equity fund designed to mimic the performance of the Russell 2000 Index. Based upon the information provided below, what is the best estimate of the tracking error of Valance relative to the Russell 2000 Index?

Annual volatility of Valance:	35%
Annual volatility of the Russell 2000 Index:	40%
Correlation between Valance and the Russell 2000 Index:	0.90

- A. 3.1%
- B. 17.5%
- C. 39.6%
- D. 53.2%

Q-38. On January 1, 2006, a pension fund has assets of EUR 100 billion and is fully invested in the equity market. It has EUR 85 billion in liabilities. During 2006, the equity market declined by 15% and yields increases by 1.2%. If the modified duration of the liabilities is 12.5, what is the pension fund's surplus on December 31, 2006?

- A. EUR 15.00 billion
- B. EUR 12.93 billion
- C. EUR 12.75 billion
- D. EUR 12.57 billion

Q-39. At the end of 2007, Chad & Co.'s pension had USD 350 million worth of assets that were fully invested in equities and USD 180 million in fixed-income liabilities with a modified duration of 14. In 2008, the wide spread effects of the subprime crisis hit the pension fund, causing its investment in equities to loss 50% of their market value. In addition, the immediate response from the government – cutting interest rates – to salvage the situation, caused bond yields to decline by 2%. What was the change in the pension fund's surplus in 2008?

- A. USD -55.4 million
- B. USD -124.6 million
- C. USD -225.4 million
- D. USD -230.4 million

Q-40. An analyst reports the following fund information to the advisor of a pension fund that currently invests in government and corporate bonds and carries a surplus of USD 10 million.

Pension Assets	Assets	Liabilities
Amount (in USD million)	100	90
Expected Annual Growth	6%	7%
Modified Duration	12	10
Annual Volatility of Growth	10%	5%

To evaluate the sufficiency of the fund's surplus, the advisor estimates the possible surplus values at the end of one year. The advisor assumes that annual returns on assets and the annual growth of the liabilities are jointly normally distributed and their correlation coefficient is 0.8. The advisor can report that, with a confidence level of 95%, the surplus value will be greater than or equal to:

- A. USD -11.4 million
- B. USD -8.3 million
- C. USD -1.7 million
- D. USD 0 million

Q-41. Which of the following statements about risk management in the pension fund industry is correct?

- A. A pension plan's total VaR is equal to the sum of its policy-mix VaR and active management VaR.
- B. Pension fund risk analysis does not consider performance relative to a benchmark.
- C. In most defined-benefit pension plans, if liabilities exceed assets, the shortfall does not create a risk for the plan sponsor.
- D. From the plan sponsor's perspective, nominal pension obligations are similar to a short position in a bond.

Q-42. A company's pension fund is established as a defined benefit plan, and therefore the board must consider funding risk. Which of the following statements about the pension fund's funding risk is correct?

- A. The longer the horizon for expected payouts, the lower the funding risk.
- B. Decreases in interest rates will reduce funding risk.
- C. The funding risk has been effectively transferred to the employees.
- D. Funding risk represents the true long-term risk to the plan sponsor.

Q-43. The AT&T pension fund has an allocation of \$60 million devoted to U.S. equities. Now the fund wants to allocate this \$60 million to two managers. This is equivalent to a risk budget of \$3.948 million. Each manager has a TEV of 6%. The fund managers have different capabilities, their IRs are 0.6(manager 1) and 0.4(manager 2). To achieve an exact TEV of 4% and information ratio of 0.725, the weight for each should be?

- A. Manager1: 55.17%, Manager2: 36.78%
- B. Manager1: 55.17%, Manager2: 44.83%
- C. Manager1: 36.78%, Manager2: 55.17%
- D. Insufficient information to calculate

Q-44. A portfolio manager at a pension fund is presenting on investment strategies during a training for newly-hired portfolio analysts. The manager discusses low volatility strategies, illustrates historical performance measures of firms that apply these strategies, and draws attention to the benchmarks used. Which of the following statements about low volatility strategies would be correct for the manager to make during the presentation?

- A. The strategies tend to have significant alphas relative to standard market capitalization benchmarks.
- B. The strategies tend to have negative alphas relative to dynamic factors such as value or momentum.
- C. The strategies tend to generate high alphas over the risk-free rate but negligible alphas over any other benchmark.
- D. The strategies tend to generate low alphas if the benchmark used is adjusted for risk and high alphas otherwise.

Q-45. A large pension fund requires that the fund's managers do not breach the 2% tracking error limit at any point in time. A fund manager's performance for the most recent period is summarized below:

Average return: 2.8%

Volatility of returns: 1.9%

Average return in excess of the benchmark (average active return): 0.6%

Volatility of active returns: 1.7%

If the current risk-free interest rate is 1.2%, which of the following is correct

- A. The manager's average active return is below 2%, therefore the manager breached the limit.
- B. The manager's average return in excess of the risk-free interest rate is below 2%, therefore the manager breached the limit.
- C. The volatility of active returns achieved by the manager is below 2%, therefore the manager did not breach the limit.
- D. The volatility of returns achieved by the manager is below 2%, therefore the manager did not breach the limit.

Q-46. An investment fund uses risk budgeting as part of its risk management process. Risk is calculated and monitored using delta-normal VaR at the 99% confidence level. The fund's total principal of EUR 100 million is invested across four asset classes comprised of European stocks, non-European stocks, European bonds, and non-European bonds. The

total volatility profile of the fund is maintained at 5%. Information on the four asset classes is given below:

	Asset classes	Weight	Average return	Volatility	Correlation			
					1	2	3	4
1	European stocks	40%	12.99%	9.25%	1.00	-0.05	-0.07	-0.03
2	Non-European stocks	12%	10.82%	11.91%	-0.05	1.00	0.02	-0.01
3	European bonds	22%	5.10%	5.76%	-0.07	0.02	1.00	0.02
4	Non-Europea bonds	26%	10.53%	11.94%	-0.03	-0.01	0.02	1.00

What is the sum of the risk budgets that should be allocated to the four asset classes?

- A. EUR 11.64 million
- B. EUR 22.12 million
- C. EUR 38.86 million
- D. EUR 100.0 million

Q-47. A group of risk managers in a newly established asset management firm is assigned to implement the risk management process that includes three fundamental dimensions: risk planning, risk budgeting and risk monitoring. The managers start by discussing the components of and the guidelines included in the risk plan. Which of the following statements is correct?

- A. Qualitative scenario analyses can be incorporated into a risk plan to identify factors that can cause aspects of the risk plan to fail.
- B. A risk plan can set volatility goals but cannot incorporate the effects of the organization's key dependencies on these goals.
- C. Extreme events should not be included in an organization's risk plan but they should be included in its strategic plan.
- D. A risk plan should include an estimate of return on equity found by using the return on risk capital for each allocation taken independently.

Q-48. A fund of funds (FOF) manager is conducting a performance attribution analysis for a portfolio consisting of equity and fixed-income securities to evaluate the effects of the portfolio manager's asset allocation and security selection decisions. The FOF manager uses the information presented below:

Asset class	Portfolio weight	Benchmark weight	Portfolio return	Benchmark return
Equity	58%	50%	8%	11%

Fixed income	42%	50%	6%	7%
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What is the contribution of the portfolio manager's asset allocation decision to the portfolio's overall excess return?

- A. -2.16%
- B. -1.84%
- C. -0.16%
- D. 0.32%

Q-49. A pension fund manager is planning to invest a portion of the fund's portfolio in hedge funds. The manager is concerned about the potential asymmetry in risk sharing that may occur with hedge fund investments. What action should the pension fund manager take to mitigate this risk?

- A. Allocate the money across several different hedge fund strategies to diversify away the asymmetry in risk sharing.
- B. Choose a reputable hedge fund manager that manages investments for other major pension funds.
- C. Ensure that the hedge fund managers have a sizable amount of their own wealth invested in their fund.
- D. Require the hedge fund to provide a daily position report to better monitor the potential asymmetry in risk sharing.

4.7. Key Point: Performance Measurement and Evaluation

4.7.1. 重要知识点

4.7.1.1. Portfolio Performance Measurement

➤ Performance Analysis

$$SR = \frac{E(R_P) - R_f}{\sigma_P}$$

$$TR = \frac{E(R_P) - R_f}{\beta_P}$$

$$\alpha_P = E(R_P) - \{R_F + \beta_P[E(R_M) - R_F]\}$$

$$IR = \frac{E(R_P) - E(R_B)}{\sigma_{TE}}$$

➤ Performance Attribution

- Refers to the set of techniques used by performance analysts to identify the sources of value addition to the portfolio. For example, how much of the performance (excess returns above benchmark) is attributable to the selection of the right asset classes or how much is attributable to selection of right sector or security within an asset class.

4.7.2. 基础题

Q-50. A manager who obtains an average alpha of 2.5% with a tracking-error of 4%. If he wish the result to be significant to 95%, how many years it is necessary to observe the portfolio return?

- A. 8.8 years
- B. 9.8 years
- C. 10.8 years
- D. 11.8 years

Q-51. Based on 60 monthly returns, you estimate an actively managed portfolio alpha = 1.24% and standard error of alpha = 0.1278%. The portfolio manager wants to get due credit for producing positive alpha and believes that the probability of observing such a large alpha by chance is only 1%. Calculate the t-statistic, and based on the estimated t-value would you accept (or reject) the claim made by the portfolio manager.

- A. $t = 9.70$, accept
- B. $t = 2.66$, accept
- C. $t = 2.66$, reject
- D. $t = 9.70$, reject

Q-52. Consider the following performance data for a sample period:

	Portfolio (P)	Market (M)
Average return	15%	9%
Beta	1.6	1.0
Standard deviation	32%	24%
Tracking error	20%	0
Risk free rate	—	3%

If the Portfolio (P) is one sub-portfolio that is combined with several other portfolios into a large investment fund, which is the appropriate risk-adjusted performance measure (RAPM) and what is its value for Portfolio (P)?

- A. Sharpe of 25.0%
- B. Treynor of 6.0%
- C. Treynor of 7.5%
- D. Information ratio of 12.0%

Q-53. Rick Masler is considering the performance of the managers of two funds, the HCM Fund

and the GRT Fund. He uses a linear regression of each manager's excess returns (r_i) against the excess returns of a peer group (r_B):

$$r_i = a_i + b_i \times r_B + e_i$$

The information he compiles is as follows:

Fund	Initial Equity	Borrowed Funds	Total Investment Pool	a_i	b_i
HCM	USD 100	USD 0	USD 100	0.0150	0.9500
				(t = 4.40)	(t = 12.1)
GRT	USD 500	USD 3,000	USD 3,500	0.0025	3.4500
				(t = 0.002)	(t = 10.20)

Based on this information, which of the following statements is correct?

- A. The regression suggests that both managers have greater skill than the peer group.
- B. The a_i term measures the extent to which the manager employs greater or lesser amounts of leverage than do his/her peers.
- C. If the GRT Fund were to lose 10% in the next period, the return on equity (ROE) would be -60%.
- D. The sensitivity of the GRT fund to the benchmark return is much higher than that of the HCM fund.

Q-54. A portfolio has an average return over the last year of 13.2%. Its benchmark has provided an average return over the same period of 12.3%. The portfolio's standard deviation is 15.3%, its beta is 1.15, its tracking error volatility is 6.5% and its semi-standard deviation is 9.4%. Lastly, the risk-free rate is 4.5%. Calculate the portfolio's information Ratio (IR).

- A. 0.569
- B. 0.076
- C. 0.138
- D. 0.096

Q-55. Portfolio Q has a beta of 0.7 and an expected return of 12.8%. The market risk premium is 5.25%. The risk-free rate is 4.85%. Calculate Jensen's Alpha measure for Portfolio Q.

- A. 7.67%
- B. 2.70%
- C. 5.73%
- D. 4.27%

Q-56. A risk manager runs a performance attribution analysis on an actively managed portfolio

using a selected benchmark. The weights and performance of the different market sectors within the portfolio and the benchmark are given below:

	Benchmark		Portfolio	
Market Sector	Weight	Annual Return	Weight	Annual Return
Equity	20%	8%	40%	6%
Fixed Income	50%	4%	55%	5%
Cash	30%	2%	5%	3%

What conclusion can be drawn from the data above by using common performance attribution analysis?

- A. The portfolio outperforms the benchmark primarily because of the contribution of asset allocation.
- B. The portfolio outperforms the benchmark primarily because of the contribution of security selection within market sectors.
- C. The portfolio underperforms the benchmark primarily because of the contribution of asset allocation.
- D. The portfolio underperforms the benchmark primarily because of the contribution of security selection within market sectors.

Q-57. An individual investor reviews historical data on the performance of several investment funds and decides to create a USD 1 million portfolio that mimics the strategy of Fund CRN, which has consistently generated high alphas. The investor gathers Fund CRN's monthly returns over the last 10 years and regresses Fund CRN's monthly excess returns over the risk-free rate against the Fama-French model's three factors as well as a momentum factor. The investor obtains the following statistically significant estimates:

	Coefficient	T-statistic
Alpha	0.08	2.10
MKT loading	0.55	6.72
SMB loading	-0.63	4.35
HML loading	0.36	3.20
UMD loading	-0.07	2.77
Adj R2	0.56	

Which of the following positions is a component of the mimicking portfolio?

- A. USD 7,000 long position in stocks showing positive momentum
- B. USD 360,000 long position in value stocks
- C. USD 450,000 short position in T-bills
- D. USD 630,000 short position in growth stocks

Q-58. A risk analyst is evaluating an investment portfolio using the Fama-French three-factor

model. The analyst regresses thirty years of weekly portfolio returns against the three factors of the model. The analyst obtains the following regression results:

Alpha	0.10
Market coefficient	0.52
SMB coefficient	0.18
HML coefficient	-0.70

Assuming all estimated coefficients are statistically significant, which of the following is correct?

- A. There is a positive correlation between portfolio return and the size factor, which indicates that the portfolio moves together with large-cap stocks.
- B. There is a positive correlation between portfolio return and the value factor, which indicates that the portfolio moves together with growth stocks.
- C. There is a negative correlation between portfolio return and the size factor, which indicates that the portfolio moves together with large-cap stocks.
- D. There is a negative correlation between portfolio return and the value factor, which indicates that the portfolio moves together with growth stocks.

Q-59. A risk analyst at a bank is asked to prepare a report that tracks the relationship between volatility and asset performance. The analyst assesses the performance of various asset classes using empirical evidence over the last three decades and compares returns on those asset classes with changes in market volatility. Which of the following would be a correct statement for the analyst to include in the report?

- A. Currency strategies such as currency carry trades tend to perform poorly during periods of high volatility.
- B. When volatility is rising, all assets are either positively or negatively affected, with the exception of risk-free bonds.
- C. Whether the relationship between stock returns and volatility is positive or negative depends on the phase of the business cycle.
- D. When volatility is rising, stock returns tend to increase but bond returns tend to decrease.

Q-60. A money manager who has recently received a small amount of new capital is planning to invest this capital into an existing fund, which is benchmarked to an index. Rather than investing in a new asset to be included in the fund, the manager is planning to increase the holding of one of the fund's four assets. Information about these assets, and their performances during the most recent evaluation period, are given below:

Asset	Portfolio weight	Return	Volatility of return	Beta to the portfolio
BDE	0.35	14%	19%	1.20
JKL	0.30	13%	18%	0.90
MNO	0.25	13%	16%	1.00
STU	0.10	10%	10%	0.80

The portfolio manager wants to select the asset that has the lowest marginal VaR as long as its

Jensen's alpha is greater than or equal to the market risk premium. Assuming the risk-free interest rate is 3% and the market return is 8%, which asset should the portfolio manager select?

- A. Asset BDE
- B. Asset JKL
- C. Asset MNO
- D. Asset STU

Q-61. A newly established pension fund hires a risk consultant to help develop its risk management framework and tools. During an initial meeting with the fund managers, the consultant discusses the fundamental dimensions of the risk management process and describes risk planning, risk budgeting, and risk monitoring, as well as how each of these three dimensions should be structured. Which of the following statements about the risk plan or the risk budget is correct?

- A. The risk budget should define acceptable levels of return on risk capital (RORC) for each risk capital allocation.
- B. The risk budget should identify the firm's critical dependencies regarding funding and investment performance.
- C. The risk plan should state exactly how risk capital should be allocated among asset classes.
- D. The risk plan should set volatility goals such as VaR or tracking error for relevant time periods.

Q-62. An investment management firm is in the process of strengthening its internal control environment and forms an independent risk management unit (RMU). A group of managers at the firm is asked to prepare a summary document describing the responsibilities of the new unit. While documenting the duties of the RMU, which of the following actions would be appropriate for the managers to include?

- A. Identify appropriate software and research reports to help make trading decisions.
- B. Generate VaR levels that are consistent with the targets set in the risk plan.
- C. Oversee model analysts while they establish asset valuation models.
- D. Assess the quality of models used to measure risk.

Q-63. An investor is comparing the performances of two portfolio managers who have been allocated an equal amount of investment funds. The managers apply the same strategy with the same constraints, and their portfolios are not diversified. The investor gathers the following data about the two managers and the market index:

	Manager 1	Manager 2	Market index
Average return	32%	28%	22%
Beta with respect to market index	1.2	1.4	1.0
Standard deviation of returns	18%	14%	10%

Tracking error	8%	6%	0%
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The risk-free rate of interest is 3%. Which of the following is an appropriate measure to use and the correct conclusion to reach when comparing the performances of the two managers?

- A. The Modigliani-squared measure, which shows that Manager 1 outperforms Manager 2 by 2%
- B. The Modigliani-squared measure, which shows that Manager 2 outperforms Manager 1 by 2%
- C. Treynor's measure, which shows that Manager 1 outperforms Manager 2 by 6%
- D. Treynor's measure, which shows that Manager 2 outperforms Manager 1 by 6%

4.8. Key Point: Hedge Funds

4.8.1. 重要知识点

4.8.1.1. Hedge Fund Trading Strategy

- 1) Equity long/short strategy: go long and short similar securities to exploit mispricing-decreases market risk and generates alpha.
- 2) Global macro strategy: makes leveraged bets on anticipated price movements in broad equity and fixed-income markets, interest rates, foreign exchange, and commodities.
- 3) Emerging markets strategy: invests in developing countries' securities or sovereign debt.
- 4) Fixed-income arbitrage strategy: long/short strategy that looks for pricing inefficiencies between various fixed-income securities.
- 5) Convertible arbitrage strategy: investor purchases a convertible bond and sells the underlying stock.
- 6) Merger arbitrage strategy: involves purchasing shares in a target firm and selling short shares in the purchasing firm.
- 7) Distressed investing strategy: purchase bonds of distressed company and sell short the stock, anticipating that the shares will eventually be worthless.

4.8.1.2. Hedge Fund Trading Strategy

- Fund of hedge funds: perform screening and due diligence of other funds. Fees can be extensive, and the due diligence does not always identify fraud. A key advantage is diversification benefit without large capital commitment.

4.8.1.3. Hedge Fund Risk

- Style Drift
 - Changes in the risk factor exposures
 - Changes in the overall risk of the fund

4.8.2. 基础题

Q-64. Which of the following statements about convertible arbitrage hedge fund strategies is

correct?

- A. Credit risk plays only a minor role in convertible arbitrage hedge funds.
- B. Investing in convertible arbitrage does not require an understanding of liquidity considerations as the market for convertible securities is sufficiently liquid today.
- C. Gamma trading entails significant directional exposure to the equity markets.
- D. Re-hedging after significant moves of the underlying stock price is the essence of gamma trading.

Q-65. Identify the risks in a convertible arbitrage strategy that takes long positions in convertible bonds hedged with short positions in Treasuries and the underlying stock.

- A. Short implied volatility
- B. Long duration
- C. Long stock delta
- D. Positive gamma

Q-66. George Smith, a hedge fund manager, has just established a short position in short-term Swiss government bonds that are currently yielding 3.5% and a long position in short-term Italian government bonds that are yielding 4.2%. Smith believes the market has underestimated the probability that the Swiss Franc will appreciate relative to the euro. Which of the following hedge fund strategies is most similar to Smith's strategy?

- A. Pair trading strategy.
- B. Managed futures strategy.
- C. Global macro strategy.
- D. Event-driven strategy.

Q-67. A fund of hedge funds combines a mix of strategy sectors, managers, and styles, and therefore fund of funds risk managers need to understand the common attributes of hedge fund strategies. Which of the following statements is incorrect?

- A. Equity market neutral funds aim to generate returns that have low correlation with the overall equity market and to insulate their portfolios from broad market risk factors.
- B. Convertible arbitrage funds typically purchase securities that are convertible into the issuer's stock and simultaneously short the underlying stock. These funds earn returns in part from gamma trading on the stock's volatility.
- C. Merger arbitrage funds buy the stock of an acquisition target company and simultaneously short the bidding company's stock. These funds have large exposure to deal risk.
- D. Equity short-selling funds sell stocks not currently owned by the seller in order to take a

directional bet that the stock price will decline. These funds tend to be uncorrelated with traditional long-only equity portfolios.

Q-68. Every year Business Week reports the performance of a group of existing equity mutual funds, selected for their popularity. Taking the average performance of this group of funds will create?

- A. Survivorship bias only
- B. Selection bias only
- C. Both survivorship and selection bias
- D. Instant-history bias only

Q-69. A factor analysis of returns for hedge funds employing a equity market-neutral strategy produces strongly positive performance information for the strategy (for example, impressive Sharpe ratios). However, the analysis is guilty of neglecting the effects of survivorship bias. If the problem is survivorship bias, which of the following criticisms of the methodology is best?

- A. The sample is too small
- B. The historical window is too short
- C. Risk metrics needs to be included along with return metrics
- D. Past performance is no guarantee of future performance

Q-70. Which of the following statements are true?

- I. Hedge fund manager compensation is often symmetric (i.e., a dollar of gain has the opposite impact on compensation as a dollar of loss), while the compensation of mutual fund managers is almost always asymmetric.
- II. Leverage obtained through lines of credit increases the risk of a hedge fund more than leverage obtained by issuing debt, because unexpected cancellation of a line of credit by a lender during troubled times can force a fund to liquidate its positions in illiquid markets.
- III. A hedge fund investor should pay performance-based compensation to the manager for producing alpha, but should not pay performance-based compensation to a hedge fund manager who has done well because the fund invests in risk factors that mirror the performance of his style or strategy, and the style or strategy has performed well.
- IV. The lack of hedge fund transparency is particularly Problematic for investors with fiduciary responsibilities such as pension fund managers, and to secure funding from these investors, hedge fund managers often have to provide more information

to these investors.

- A. I, II, and IV only.
- B. II, III, and IV only.
- C. II and IV only.
- D. I and III only.

Q-71. The beneficiary of a trust who reaches adulthood and inherits assets that are mostly liquid meets with an investment advisor to consult on various investment options. During the meeting, the beneficiary expresses an interest in understanding the different types of funds, especially hedge funds. Which of the following statements, if made by the investment advisor, is correct?

- A. Market timing skills of indexed fund managers are more important than market timing skills of hedge fund managers.
- B. Demands of institutional investors regarding the privacy of their investments have caused hedge funds to become less transparent over time.
- C. Leveraging of investor capital and shorting securities are practices that are used more extensively by hedge funds than by most conventional funds.
- D. Hedge funds must disclose their investment strategies to existing and prospective clients but exchange-traded funds do not have to.

Q-72. A portfolio analyst is studying the history of hedge funds. While reviewing the milestones that shaped the hedge fund industry, the analyst focuses on institutional investors replacing private individuals as the dominant hedge fund investor group in the early 2000s. The analyst examines the effects of this major development on the industry, including changes in investor expectations, assets under management (AUM), and the level and volatility of returns. Which of the following statements is correct regarding the emergence of institutional investors as the dominant investor group in the hedge fund industry?

- A. Institutional investors have uniform expectations, which caused strategy benchmarking based on peer group alphas to become more effective.
- B. Institutional investors allocated their investments evenly among small, medium, and large hedge funds, which decreased the concentration of AUM in the hedge fund industry.
- C. Some hedge funds started to pursue more conservative strategies and hedge fund indices started to generate lower returns than the S&P 500 Index.
- D. Some hedge funds changed their strategies, but the volatility of hedge fund index returns continued to be lower than that of the S&P 500 Index returns.

- Q-73.** Which of the following is the best summary of the primary principal-agent risk (or problem) in the hedge fund industry, according to Fung and Hsieh?
- A. Although "clearly of academic interest," there is neither credible evidence nor a plausible narrative in support of a realistic principle-agent problem in the hedge fund industry
 - B. The principle-agent risk arises due to the incentive fee structure and the high water mark (HWM) but can be mitigated by agents (fund managers) who invest a sizable amount of their own wealth in the fund
 - C. The principle-agent risk arises due to the fact that hedge fund managers have more information about the fund's investments than managers, and there is no easy solution to this principle-agent problem because the principals are not full-time investors themselves
 - D. The principle-agent risk arises due to the fact that the hedge fund manager, who charges a management fee as a percentage of assets under management (AUM), wants to grow the size of the firm rather than outperform, but this can be addressed simply by avoiding big funds in favor of small funds

- Q-74.** An individual investor wants to invest USD 8 million in exchange-traded funds (ETFs) or private equity funds (PEFs). The investor obtains the previous year's returns for several ETFs and calculates summary statistics such as volatility and correlation based on these returns. The investor also reviews a database of reported returns and volatilities for several PEFs and then selects two potential investments in each asset class. Using the data from the sources described above, the investor generates the following information for the four potential investments:

Asset	1-year return	Annual volatility of returns
Broad equity market index ETF (ETF1)	6.5%	11.4%
Growth stock ETF (ETF2)	8.3%	13.6%
Private equity fund 1 (PEF1)	7.4%	12.3%
Private equity fund 2 (PEF2)	10.2%	11.1%

Correlation of returns between ETF1 and ETF2	0.67
Correlation of returns between ETF1 and PEF1	0.25
Correlation of returns between PEF1 and PEF2	0.41

The manager evaluates this information while also considering the potential biases and uncertainties in the reported data. Which of the following conclusions is most appropriate for the investor to make?

- A. The correlation of returns between PEF1 and ETF1 is more accurate than the correlation

of returns between ETF1 and ETF2.

- B. The entire USD 8 million should be allocated to PEF2 because it is clearly the superior investment from a return to risk perspective.
- C. The number of assets in PEFs are typically higher than the number of assets in ETFs, which makes PEFs much less risky than ETFs over longer time horizons.
- D. Summary statistics computed using reported returns of PEFs can be biased downward, which compromises the reliability of these risk measures.

4.9. Key Point: Due Diligence and Fraud Risk

4.9.1. 重要知识点

4.9.1.1. Detecting Fraud

- Efficacy of Information Disclosures
 - Past Regulatory Violations
 - Conflicts of Interest
 - Monitoring
- Barriers and Costs
- Improve Investors' Ability

4.9.2. 基础题

Q-75. A due diligence specialist is evaluating the risk management process of a hedge fund in which his company is considering making an investment. Which of the following statements best describes criteria used for such an evaluation?

- A. Because of the overwhelming importance of tail risk, the company should not invest in the fund unless it fully accounts for fat tail using extreme value theory at the 99.99% level when estimating VaR.
- B. Today's best practices in risk management require that a fund employ independent risk service providers and that these service providers play important roles in risk-related decisions.
- C. When considering a leveraged fund, the specialist should assess how the fund estimates risks related to leverage, including funding liquidity risks during periods of market stress.
- D. It is crucial to assess the fund's valuation policy, and in general if more than 10% of asset prices are based on model prices or broker quotes, the specialist should recommend against investment in the fund regardless of other information available about the fund.

Q-76. In performing due diligence on a potential investment manager, which of the following factors is the least important for the investor to consider?

- A. Risk controls
- B. Business model

- C. Past performance
- D. Investment process

Q-77. A due diligence specialist at an asset management firm is evaluating the risk management process of a hedge fund in which the firm is considering making an investment. Which of the following statements best describes appropriate criteria the specialist should use for such an evaluation?

- A. The firm should ensure that the hedge fund allows direct, in-person communications with the fund's senior management or key decision makers at the fund.
- B. Today's best practices in risk management require that a fund employ independent risk service providers and that these service providers play important roles in risk-related decisions.
- C. When considering investing in a leveraged fund, the company should not invest in the fund unless the fund's gross leverage ratio is above the peer group average.
- D. It is crucial to assess the fund's valuation policy, and in general if more than 10% of asset prices are based on model prices or broker quotes, the specialist should recommend against investment in the fund.

Q-78. You are tasked to perform due diligence on a hedge fund and specifically its risk management process. According to Mirable, which of the following is MOST LIKELY to be a red flag?

- A. The fund's Risk Manager reports to the Portfolio Managers with dotted-line reporting to the Traders.
- B. The fund employs an independent Risk Service Provider but the provider is not co-located at the firm's headquarters.
- C. The fund claims that its process for originating and controlling risk is "complex".
- D. The fund employs different strategies but they each tend to incur a different set of risks, rather than a common set of risks.

Q-79. Each of the following is least likely a potential indicator of hedge fund Fraud risk?

- A. A high percentage of illiquid investments or those marked to market by the manager.
- B. Personal trading by managers in the same securities or similar securities as the fund.
- C. Fixed operating costs are above the industry average while manager compensation has a prominent variable component.
- D. Unusually strong performance claims; e.g., hedge fund performance claims are better than market average over a long period of time.

- Q-80.** According to Mirable, which of the following is TRUE about hedge fund due diligence?
- A. An investor must use a comprehensive checklist to ensure that nothing is left out or omitted yet remain free to ask open-ended questions that provide insights into a firm's philosophy or culture.
 - B. Due diligence is an activity with diminishing marginal returns which consumes the time of the fund's investment professions; therefore a potential investor is wise to limit the inquiry to a quantitative analysis which tends to be objective and fact-based.
 - C. Hedge fund are still largely unregulated and, unfortunately, anyone who feels that have been defrauded by a hedge fund cannot realistically report to a government agency as no single agency has jurisdiction.
 - D. Because due diligence is focused on the firm's risk management process, fund operating environment, business model and fraud risk, the investment strategy (e.g., equities, fixed-income, distressed) is largely irrelevant.
- Q-81.** A consultant is reviewing the disclosures hedge funds are required to file with the US Security and Exchange Commission. The consultant highlights how investors can use these disclosures to identify and manage fraud risk. Which of the following statements is correct?
- A. Disclosures of past regulatory or legal violations are not effective in predicting fraud because circumstances almost always change.
 - B. There is not enough evidence to conclude that investors of hedge funds are compensated for fraud risk by receiving higher returns.
 - C. Regulatory and legal prohibitions prevent fraud, and insufficient requirements on disclosure of violations cause fraud.
 - D. Regulatory and legal violation disclosures benefit only a small fraction of investors who have access to high-cost contemporaneous data.

Solutions

4. Risk Management and Investment Management

Q-1. Solution: A

Assets, including corporate bonds, private equity, and hedge funds, are not considered factors themselves, but contain many factors, such as equity risk, interest rate risk, volatility risk, and default risk.

Some assets, like equities and government bonds, can be thought of as factors themselves. Factors may also include the market (a tradable investment factor), interest rates, or investing styles (including value/growth, low volatility, or momentum).

Q-2. Solution: D

There are three similarities between food and assets:

1. Factors matter, not assets.
2. Assets are bundles of factors.
3. Different investors need different risk factors.

Q-3. Solution: B

Here is a summary of the successes of CAPM ("ideas it gets right"):

1. Don't hold an individual asset, hold the factor.
2. Each investor has his own optimal exposure of factor risk.
3. The average investor holds the market.
4. The factor risk premium has an economic story.
5. Risk is factor exposure.
6. Assets paying off in bad times have low risk premiums.

Q-4. Solution: A

The CAPM does not assume uniform taxes and transaction costs; it assumes there are no taxes or transaction costs (i.e., frictionless markets). The other limiting assumptions of the CAPM include:

1. Investors only have financial wealth.
2. Investors have mean-variance utility.
3. Investors have a single period investment horizon.
4. Investors have homogeneous (identical) expectations.
5. All investors are price takers.

Q-5. Solution: A

Assets that have losses during periods of low market returns have high betas (high sensitivity to market movements), which indicates they are risky and, therefore, should have high risk premiums. Low beta assets have positive payoffs when the market performs poorly, making them valuable to

investors. As a result, investors do not require high risk premiums to hold these assets.

Q-6. Solution: D

The higher the expected payoff of an asset in bad times, the lower the assets expected return. Assets that have a positive payoff in bad times are valuable to hold, leading to high prices and, therefore, low expected returns.

Q-7. Solution: C

A company's book value per share is equal to total assets minus total liabilities all divided by shares outstanding. It indicates, on a per-share basis, what a company would be worth if it liquidated its assets and paid off its liabilities. Value stocks have high book-to-market ratios while growth stocks have low book-to-market ratios.

Q-8. Solution: D

Momentum is a positive feedback strategy

Momentum is observed in every asset class: we observe it in international equities, commodities, government bonds, corporate bonds, industries and sectors, and real estate.

Momentum returns are not the opposite of value returns: Value is a negative feedback strategy, where stocks with declining prices eventually fall far enough that they become value stocks. Then value investors buy them when they have fallen enough to have attractive high expected returns. Value investing is inherently stabilizing. Stocks with high past returns are attractive, momentum investors continue buying them, and they continue to go up! Positive feedback strategies are ultimately destabilizing and are thus subject to periodic crashes.

Q-9. Solution: B

The Fama-French model includes the following three risk factors:

1. The traditional capital asset pricing model market risk factor.
2. A factor that captures the size effect (SMB).
3. A factor that captures the value/growth effect (HML).

The winners minus losers (WML) momentum factor was discovered by Jagadeesh and Titman in 1993 and was first put into use by Carhard to build a four-factor model.

Q-10. Solution: C

Value and momentum are opposite each other in that value investing is inherently stabilizing. It is a negative feedback strategy where stocks that have fallen in value eventually are priced low enough to become value investments, pushing prices back up. Momentum is inherently destabilizing. It is a positive feedback strategy where stocks that have been increasing in value are

attractive to investors, so investors buy them, and prices increase even more. Momentum can be riskier than value or size investing in that it is more prone to crashes.

Q-11. Solution: B

The low-risk anomaly violates the CAPM and suggests that low beta stocks will outperform high-beta stocks. This has been empirically proven with several studies. The CAPM points to a positive relationship between risk and reward, but the low-risk anomaly suggests an inverse relationship.

Q-12. Solution: C

An appropriate benchmark should be well-defined, replicable, tradeable, and risk-adjusted. If the benchmark is not on the same risk scale as the assets under review, then there is an unfair comparison.

Q-13. Solution: B

Grinold's fundamental law of active management focuses on the trade-off of high quality predictions relative to placing a large number of investment bets. Investors can focus on either action to maximize their information ratio, which is a measure of risk-adjusted performance. While sector allocation is a very important component of the asset allocation decision, Grinold focused only on the quality of predictions and the number of investment bets made.

Q-14. Solution: D

An investor should consider adding multiple factors to the regression analysis to potentially improve the adjusted R^2 measurement, potentially increase the tests of statistical significance, and to search for a benchmark that is more representative of a portfolio's investment style.

Q-15. Solution: A

Potential explanations for the risk anomaly include: the preferences of investors, leverage constraints on retail investors that drive them to buy pre-leveraged investments in the form of high-beta stocks, and institutional investor constraints like prohibitions against short selling and tracking error tolerance bands.

Q-16. Solution : A

BR is the number of bets per year.

Q-17. Solution: A

Quadratic programming requires many more inputs than other portfolio construction techniques because it entails estimating volatilities and pair-wise correlations between all assets in a portfolio.

Quadratic programming is a powerful process, but given the large number of inputs it introduces the potential for noise and poor calibration given the less than perfect nature of most data.

On the other hand, the screening technique strives for risk control by including a sufficient number of stocks that meet the screening parameters and by weighting them to avoid concentrations in any particular stock. However, screening does not necessarily select stocks evenly across sectors and can ignore entire sectors or classes of stocks entirely if they do not pass the screen. Therefore, risk control in a screening process is fragmentary at best.

Stratification separates stocks into categories (for example, economic sectors) and implements risk control by ensuring that the weighting in each sector matches the benchmark weighting. Therefore, it does not allow for overweighting or underweighting specific categories.

Linear programming does not necessarily select the portfolio with the lowest level of active risk. Rather, it attempts to improve on stratification by introducing many more dimensions of risk control and ensuring that the portfolio approximates the benchmark for all these dimensions.

Q-18. Solution: B

We can remove biases or undesirable bets from our alphas by neutralization. Benchmark neutralization means that the benchmark has 0 alpha. If our initial alphas imply an alpha for the benchmark, the neutralization process recenters the alphas to remove the benchmark alpha. We may also want to make the alphas cash-neutral; i.e., the alphas will not lead to any active cash position. It is possible to make the alphas both cash- and benchmark-neutral. We can also neutralize the alphas against the risk factors. The neutralized alphas will include only information on the factors the manager can forecast, along with specific asset information. Once neutralized, the alphas of the risk factors will be 0. For example, a manager can ensure that her portfolios contain no active bets on industries or on a size factor.

A is incorrect. Stratification is not an alpha refinement method; it is a portfolio construction method.

C is incorrect. Alphas must have already been scaled when the constraints were incorporated into the calculations.

D is incorrect. Analysts already applied trimming by excluding very large positive and negative alphas from calculations.

Q-19. Solution: A

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Q-20. Solution: A

Optimal Risk Aversion Coefficient = $IR/(2TE) = 0.8/(2 \cdot 12) = 0.033$

The optimal risk aversion coefficient is 0.033.

Q-21. Solution: D

D is correct. The first step is to calculate the VaR of the original portfolio of two equities, U.S (u) and emerging markets (e). This can be derived by using the following equation:

$VaR_p = \sqrt{VaR_u^2 + VaR_e^2 + 2\rho \times VaR_u \times VaR_e}$, where ρ is the correlation coefficient.

(i) Initial position: The portfolio 1-day 95% VaR (before the rebalancing) is therefore:

$$VaR_p = \sqrt{1.2^2 + 1.2^2 + 2 \times 0.36 \times 1.2 \times 1.2} = \text{USD } 1.979 \text{ million}$$

(ii) Rebalanced position: 1-day 95% VaR: After the rebalance, the market value of the position in the U.S. equities is reduced by $8/48 = 0.1667$, so VaR_u is now equal to $(1 - 0.1667) \times (\text{USD } 1.2 \text{ million}) = \text{USD } 1.0 \text{ million}$. Meanwhile the market value for the position in the emerging market equities has increased by $8/35 = 0.2286$ so that VaR_e is now $(1 + 0.2286) \times (\text{USD } 1.2 \text{ million}) = \text{USD } 1.474 \text{ million}$. Hence the 1-day 95% VaR of the new portfolio (after rebalancing) = USD 2.058 million and is calculated as follows:

$$VaR_p = \sqrt{1.474^2 + 1^2 + 2 \times 0.36 \times 1.474 \times 1} = \text{USD } 2.058 \text{ million}$$

(iii) Next, convert the 1-day 95% VaR to 10-day 95% VaR:

$$10\text{-day } 95\% \text{ VaR} = (1\text{-day } 95\% \text{ VaR}) \times \sqrt{10}/1 = 2.058 \times 3.162278 = \text{USD } 6.508 \text{ million.}$$

(iv) Finally, convert the 10-day 95% VaR to 10-day 99% VaR:

$$10\text{-day } 99\% \text{ VaR} = (10\text{-day } 95\% \text{ VaR}) \times (2.326/1.645) = 6.508 \times 1.4140 = \text{USD } 9.202 \text{ million.}$$

The question is to compare the original 1-day 95% VaR (USD 1.979m) to the new rebalanced 10-day 99% VaR (USD 9.202). Thus, VaR will increase by $(9.202 - 1.979)$ million, or USD 7.223 million.

Thus, D is correct.

Q-22. Solution: B

$$\sigma_p = \sqrt{\omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + 2\rho \omega_1 \omega_2 \sigma_1 \sigma_2}$$

$$\sigma_{\text{previous}}^{1\text{-year}} = \sqrt{\left(\frac{2}{3}\right)^2 \times (25\%)^2 + \left(\frac{1}{3}\right)^2 \times (20\%)^2 + 2 \times 0.2 \times \frac{1}{3} \times \frac{2}{3} \times 20\% \times 25\%} = 0.191485$$

$$\sigma_{\text{now}}^{1\text{-year}} = \sqrt{\left(\frac{1}{3}\right)^2 \times (25\%)^2 + \left(\frac{2}{3}\right)^2 \times (20\%)^2 + 2 \times 0.2 \times \frac{1}{3} \times \frac{2}{3} \times 20\% \times 25\%} = 0.170783$$

$$\sigma_{\text{previous}}^{1\text{-year}} = \sigma_{\text{previous}}^{1\text{-day}} \times \sqrt{250} \rightarrow \sigma_{\text{previous}}^{1\text{-day}} = \frac{0.191485}{\sqrt{250}} = 0.012111$$

$$\sigma_{\text{now}}^{1\text{-year}} = \sigma_{\text{now}}^{1\text{-day}} \times \sqrt{250} \rightarrow \sigma_{\text{now}}^{1\text{-day}} = \frac{0.170783}{\sqrt{250}} = 0.010801$$

$$\text{VaR}_{\text{change}} = z \times \sigma_{\text{change}} \times P = 2.33 \times (0.012111 - 0.010801) \times 150 = 0.4578$$

The trade will decrease the VaR by 0.4578.

Q-23. Solution: C

Let $Z(99\%)$ represent the 99% confidence factor for the VaR estimate, which is 2.326, r_{XYZ} represent the correlation of stock XYZ with the portfolio, which is 0.3, and V_{XYZ} represent the value of stock XYZ, which is CAD 5 million.

$$\text{Then, } \text{MVar}_{XYZ} = Z(99\%) \times s_{XYZ} \times r_{XYZ} = 2.326 \times 15\% \times 0.3$$

$$\text{Component VaR}_{XYZ} = \text{MVar}_{XYZ} \times V_{XYZ} = \text{CAD } 523,350$$

Q-24. Solution: B

This question tests that the candidate understands correlation in calculating portfolio VaR. From the table, we can get daily volatility for each fund:

$$\text{Fund Alpha volatility: } 0.20 / 252^{0.5} = 1.260\%$$

$$\text{Fund Omega volatility: } 0.25 / 252^{0.5} = 1.575\%$$

Portfolio variance:

$$0.5^2 \times 0.01259^2 + 0.5^2 \times 0.01574^2 + 2 \times 0.5 \times 0.5 \times 0.01259 \times 0.01574 \times \rho$$

$$\text{Portfolio volatility} = (\text{portfolio variance})^{0.5}$$

$$\text{Portfolio volatility is least when } \rho = -1 \rightarrow \text{portfolio volatility} = 0.1575\%$$

$$\text{Portfolio volatility is greatest when } \rho = 1 \rightarrow \text{portfolio volatility} = 1.4175\%$$

$$\text{Therefore, 95\% VaR maximum is } 1.645 \times 0.014175 \times 1,000,000 = \text{USD } 23,316$$

Q-25. Solution: C

For uncorrelated positions, the answer is the square root of the sum of the spread VaRs:

$$\text{VaR}_p = \sqrt{10^2 + 20^2} = \$22.36 \text{ million}$$

Q-26. Solution: D

$$\begin{aligned} \text{VaR}_p &= 1.645 \times \sqrt{(1.8 \times 0.08)^2 + (3.2 \times 0.04)^2 + 2 \times 15\% \times (1.8 \times 0.08) \times (3.2 \times 0.04)} \\ &= 0.339722 \text{ Million} \end{aligned}$$

Q-27. Solution: B

An error in predicting correlation among asset classes will cause the calculation of optimal asset allocation to be in error as well. Thus, the asset allocation of the portfolio will be less than optimal. Any portfolio that does not have optimal asset allocation will, by definition, have returns that are too low for the expected level of risk. The risk of the portfolio will be overstated because the estimates of correlation among markets were too high.

Q-28. Solution: A

A is correct: The new portfolio VaR is that of asset 2 alone (USD 46.6), which implies a reduction in portfolio VaR of USD 61.6 - USD 46.6 = USD 15.0

Q-29. Solution: B

Marginal $VaR_i = \beta_i \times \text{Portfolio VaR} / \text{Portfolio Value}$

So, $\beta_i = \text{Marginal VaR}_i \times \text{Portfolio Value} / \text{Portfolio VaR}$

$$\beta_2 = 0.44 \times 200 / 61.6 = 1.429$$

Q-30. Solution: C

$VaR_p = \alpha \times \text{portfolio standard deviation} \times \text{portfolio value}$

$$= 1.645 \times 0.0592 \times \text{USD}1,000,000 = \text{USD}97,384$$

$$\text{Component } VaR_A = \text{USD}97,384 \times 0.5 \times \frac{400}{1000} = \text{USD}19,477$$

$$\text{Marginal } VaR_B = \text{USD}97,384 \times 1.2 / \text{USD}1,000,000 = 0.1169$$

Q-31. Solution: A

$$\beta = \rho \frac{\sigma_i}{\sigma_p} = 0.37 \times \frac{16\%}{21\%} = 0.2819$$

$$\text{Component VaR} = 0.2819 \times 2.326 \times 21\% \times 15\text{million} = 2.066\text{million}$$

Q-32. Solution: D

Liquidity duration is an approximation of the number of days necessary to dispose of a portfolio's holdings (of a particular share in this case) without a significant market impact. It is calculated as: $8,000 / (0.25 \times 2,000) = 16$.

Q-33. Solution: C

$$TR_A = \frac{12\% - 2\%}{1.2} = 0.0833$$

$$TR_B = \frac{10\% - 2\%}{0.7} = 0.1143$$

$$TR_C = \frac{10\% - 2\%}{0.6} = 0.1333$$

$$TR_D = \frac{(8\% - 2\%)}{0.3} = 0.2$$

Asset B, C, D have Treynor measures greater than 0.1. Of these, C has the lowest marginal VaR as its Beta to the portfolio is the lowest.

Q-34. Solution: A

We calculate the liquidity duration (LDi) of each security as shown below:

Security (i)	Qi	MDVi	Vi	LDi = Qi/(MDVi*Vi)
A	500,000	22%	522,000	4.3539
B	420,000	12%	1,328,000	2.6355
C	256,000	18%	710,000	2.0031
D	640,000	20%	848,000	3.7736

Where:

Qi: Number of shares held in security i

MDV: Maximum daily volume allowed for liquidation of security i
(expressed as a percentage of average daily volume)

Vi: Average daily volume of security i

Security A has the highest liquidity duration and will take the longest to liquidate.

Q-35. Solution: B

Optimal portfolio with the highest Sharpe ratio is attained when the ratio of expected returns to Marginal VaR is equal for all assets in a portfolio:

	TOM	JRY
	ER/MVaR	
Now	2.25	1.20
1	1.68	1.68
2	1.49	1.82
3	1.15	3.44

Q-36. Solution: B

In order to make this ratio maximized, we have:

$$\frac{\text{Position A return} - \text{risk-free rate}}{MVaR_A} = \frac{\text{Position B return} - \text{risk-free rate}}{MVaR_B}$$

Only candidate 3's results satisfy with the equation.

Q-37. Solution: B

$$\sigma_{TE}^2 = \sigma_{(R_p - R_B)}^2 = \sigma_P^2 + \sigma_B^2 - 2\rho\sigma_P\sigma_B = 0.35^2 + 0.4^2 - 2 \times 0.35 \times 0.4 \times 0.9 = 0.0305$$

$$\sigma_{TE} = 17.5\%$$

Q-38. Solution: C

The surplus at the beginning of the year was $100 - 85 = 15$ billion EUR. During the year, the equity portfolio declines 15%, or 15 billion EUR, to 85 billion EUR. Due to the increase in yields, the dollar value of the liabilities decrease by $12.5 \times 1.2\% \times 85$ billion EUR, or 12.75. Thus at the end of the year, the assets are worth $(100 - 15) = 85$ billion EUR and the liabilities $(85 - 12.75) = 72.25$ billion. The surplus is the 12.75, a decrease of 2.25 billion EUR.

Q-39. Solution: C

The change in the pension fund's surplus for the year 2008 is equal to the initial surplus S_0 at the end of 2007 less the ending surplus S_1 at the end of 2008.

The initial surplus is calculated as $S_0 = 350 - 180 = 170$.

Next we have to calculate the surplus at the end of 2008. Given the 50% decline in the equity market, the new level of assets A_1 at the end of 2008 is equal to:

$$(1 - 0.5) \times 350 = 175$$

The new level of liabilities L_1 can be calculated as:

$$L_1 = (1 - 14 \times (-0.02)) \times 180 = 230.4$$

Therefore the 2008 surplus S_1 is equal to $A_1 - L_1 = 175 - 230.4 = -55.4$ (which implies the pension fund is actually in a deficit situation at the end of 2008). The change in surplus for 2008 is hence $S_1 - S_0 = -55.4 - 170 = -225.4$ million.

Q-40. Solution: C

The lower bound of the 95% confidence interval is equal to: Expected Surplus – (95% confidence factor $\times$ Volatility of Surplus). The required variables can be calculated as follows:

$$\text{Variance of the surplus} = 100^2 \times 10\%^2 + 90^2 \times 5\%^2 - 2 \times 100 \times 90 \times 10\% \times 5\% \times 0.8 = 48.25$$

$$\text{Volatility of the surplus} = 6.94$$

$$\text{The expected surplus} = 9.7$$

$$\text{Therefore, the lower bound of the 95\% confidence interval} = 9.7 - 1.645 \times 6.94 = -1.725$$

Q-41. Solution: D

Explanation: Liabilities at a pension fund are typically composed of accumulated benefit obligations, measured by the present value of all pension benefits owed to employees discounted by an approximate interest rate. When liabilities consist mostly of nominal payments, their value in general will behave like a short position in a long-term bond.

Q-42. Solution: D

Explanation: The time horizon of payouts does not eliminate funding risk. In fact it is the mismatch between assets and liabilities that creates funding risk. In an low interest rate environment the value of equities will rise, however the value of the liabilities are likely to increase more thereby exacerbating funding risk. Funding risk is transferred to employees with a defined contribution plan. Immunizing the portfolio, essentially matching duration of assets and liabilities, will reduce funding risk.

Q-43. Solution: A

$$\omega_1 = \frac{IR_1/TEV_1}{IR_P/TEV_P} = \frac{0.6/6\%}{0.725/4\%} = 55.17\%$$

$$\omega_2 = \frac{IR_2/TEV_1}{IR_P/TEV_P} = \frac{0.4/6\%}{0.725/4\%} = 36.78\%$$

Q-44. Solution: A

A is correct. Low-risk strategies appear to have significant alpha relative to standard market capitalization benchmarks and sophisticated factor benchmarks that control for risk using dynamic value and momentum factors.

B is incorrect. See explanation for A.

C is incorrect. See explanation for A.

D is incorrect. We can't say that. Alpha is very much dependent on the benchmark used as well as whether or not that benchmark is adjusted for risk.

Q-45. Solution: C

Tracking error is the standard deviation of active returns (Active return is defined as the excess return over a benchmark). In this instance, the standard deviation of active returns is 1.7% and therefore the manager did not breach the limit.

A, B, and D are incorrect. They use incorrect criteria for determining the tracking error.

Q-46. Solution: B

Asset class	Weight (given) (A)	Volatility of returns (given) (B)	Principal by asset class (C = Ax100) (in EUR millions)	Risk budget (VaRi) (BxCx2.33) (in EUR millions)	Sum of risk budgets of asset classes
1	40%	9.25%	40	8.62	22.12

2	12%	11.91%	12	3.33	
3	22%	5.76%	22	2.96	
4	26%	11.94%	26	7.21	
Fund	100%	10.00%	100.00	22.12	11.64 < 22.12 due to diversification

A is incorrect. EUR 11.64 million is the VaR of the fund:

$$100 \times 5\% \times 2.33 = 11.64$$

C is incorrect. EUR 38.86 million is found when volatilities are multiplied by EUR 100 million and added together.

D is incorrect. EUR 100 million is the principal amount.

Q-47. Solution: A

A is correct. Scenario analyses can be incorporated into a risk plan to explore factors that can cause aspects of the risk plan to fail. These scenario analyses can be quantitative as well as qualitative.

B is incorrect. A risk plan should identify the key dependencies and incorporate the effects of their possible breakdowns on set return and volatility estimates.

C is incorrect. A company's risk plan should include low probability, severe events as well as planned strategic responses to those events.

D is incorrect. Yes, a risk plan should set an expected value for return on equity, which should be found by considering the correlations among each activity's defined minimum acceptable RORC level.

Q-48. Solution: D

The portfolio manager's asset allocation contribution is computed by summing the differences between the portfolio and benchmark weights of each asset class multiplied by the benchmark performance of that asset class.

$$\text{Contribution from asset allocation: } [(58\% - 50\%) \times 11\%] + [(42\% - 50\%) \times 7\%] = 0.32\%$$

A is incorrect. It is the contribution from security selection: $58\% \times (8\% - 11\%) + 42\% \times (6\% - 7\%) = -2.16\%$

B is incorrect. It is the total excess return: $0.32\% - 2.16\% = -1.84\%$

C is incorrect. -0.16% is found by using the below equation for each asset and summing the results: $(\text{Portfolio Weight} - \text{Benchmark Weight}) \times (\text{Portfolio Return} - \text{Benchmark Return}) = (58\% - 50\%) \times (8\% - 11\%) + (42\% - 50\%) \times (6\% - 7\%) = -0.16\%$.

Q-49. Solution: C

The risk sharing asymmetry is a situation where the hedge fund manager fully enjoys the benefits of upside risk (incentive fees), but only partially suffers from the consequences of downside risk (losing incentive fees but not losing their own capital), whereas the investor fully participates in both upside and downside risk. This asymmetry might cause a hedge fund manager to take excessive risk, especially in cases where the hedge fund is far below its high water mark so managers would need to realize a substantial gain to bring the fund back above that mark in order to begin earning incentive fees again. The most prudent approach to mitigating this risk is to ensure that the hedge fund manager has a sizable portion of their own wealth invested with the hedge fund, so that the hedge fund manager would be more conservative in their risk taking and consider

both upside and downside risk.

A is incorrect. Diversifying across many hedge funds or requiring more frequent position reporting may reduce the overall risk of investing in hedge funds but does nothing to address the risk sharing asymmetry. Managers of each fund may still be motivated to take unreasonable risks to earn incentive fees.

B is incorrect. Relying on reputation alone would not solve this risk asymmetry, as is not prudent in any case given examples of hedge fund scandals.

D is incorrect. Hedge funds normally provide very little transparency and their reports are prone to return measurement biases.

Q-50. Solution: B

$$IR \approx \frac{t_{\text{stat}}}{\sqrt{T}} \rightarrow T = \left[\frac{t_{\text{stat}}}{IR} \right]^2 = \left[\frac{1.96}{2.5\%/4\%} \right]^2 = 9.8 \text{ year}$$

Q-51. Solution: A

$$t = \frac{\alpha}{S.E(\alpha)} = \frac{1.24\%}{0.1278\%} = 9.702$$

With 60 observations and such a large t value, you would have rejected H_0 ($\alpha = 0$). The manager should receive credit for the statistically significant alpha.

Q-52. Solution: C

$$\text{Treynor}(\text{Portfolio}) = (15\% - 3\%)/1.6 = 7.5\%.$$

Q-53. Solution: D

Statement D is correct as can be seen from the b coefficient. It is higher for GRT and lower for HCM. This indicates that the sensitivity of the GRT fund to the benchmark return is much higher than that of the HCM fund.

Q-54. Solution: C

$$IR_P = \frac{13.2\% - 12.3\%}{6.5\%} = 0.138$$

Q-55. Solution: D

Jensen's alpha is defined by:

$$\alpha_P = E(R_P) - \{R_F + \beta_P[E(R_M) - R_F]\} = 0.128 - [0.0485 + 0.7 \times 0.0525] = 4.27\%$$

Q-56. Solution: A

$$R_P = 40\% \times 6\% + 55\% \times 5\% + 5\% \times 3\% = 5.3\%$$

$$R_B = 20\% \times 8\% + 50\% \times 4\% + 30\% \times 2\% = 4.2\%$$

Asset allocation: $(40\%-20\%) \times 8\% + (55\%-50\%) \times 4\% + (5\%-30\%) \times 2\% = 1.3\%$

Security selection: $(6\%-8\%) \times 40\% + (5\%-4\%) \times 55\% + (3\%-2\%) \times 5\% = -0.2\%$

Q-57. Solution: B

The mimicking portfolio should consist of (in percentages): $(1 - 0.55) = 45\%$ in T-bills + 55% in the market portfolio - 63% in small caps + 63% in large caps + 36% in value stocks - 36% in growth stocks - 7% in past winning stocks + 7% in past losing stocks. The 36% long position in value stocks translates to USD 360,000 for a portfolio of USD 1 million.

Q-58. Solution: D

The two Fama-French factors, SMB and HML, are factors that measure size and value-growth exposures, respectively. The SMB factor measures the outperformance of small-capitalization stocks versus large-capitalization stocks. A positive SML coefficient indicates that the portfolio is moving together with small cap stocks and a negative SML coefficient indicates that the portfolio is moving together with large cap stocks. The HML factor measures the outperformance of value stocks versus growth stocks. A positive HML coefficient indicates that the portfolio is moving together with value stocks and a negative HML coefficient indicates that the portfolio is moving together with growth stocks. In this question, based on the regression results, the SMB coefficient has a positive value, which means there is a positive correlation between portfolio return and SMB, and the HML coefficient has a negative value, which means there's a negative correlation between portfolio return and HML. We can rule out B and C.

A is incorrect because, as mentioned above, a positive SML coefficient indicates that the portfolio is moving together with small cap stocks.

Q-59. Solution: A

A is correct. Currency strategies fare especially poorly in times of high volatility.

B is incorrect. Risk-free bonds tend to perform well during periods of high volatility.

C is incorrect. The relationship is negative for stocks and it doesn't change sign as the economy switches from one phase to another.

D is incorrect. Stock returns also tend to decrease during periods of rising volatility.

Q-60. Solution: B

We can derive marginal VaR as:

Marginal VaR of asset $i = (VaR_p/Value_p) \times \text{Beta}_i$

Since $VaR_p/Value_p$ will be the same for all the assets, marginal VaRs of assets can be compared based on their betas.

Jensen's Alpha measure is calculated as:

Jensen's Alpha = Actual return - Expected return based on systematic risk

= Actual return - (risk-free rate + (Market return - risk-free rate) * Beta)

Note that the market risk premium = expected market return - risk-free rate = $0.08 - 0.03$

= 5%

Thus, among those assets whose Jensen's Alphas are greater than or equal to market risk premiums,

49-54

Asset JKL has the lowest Marginal VaR.

Q-61. Solution: D

D is correct. The risk plan should set expected return and volatility goals (e.g. VaR or tracking error) for relevant time periods.

A is incorrect. The risk plan should define acceptable levels of return on risk capitals (RORCs) for each risk capital allocation.

B is incorrect. The risk plan should identify critical dependencies that exist inside and outside the organization.

C is incorrect. The risk budget expresses exactly how risk capital should be allocated among asset classes and active managers.

Q-62. Solution: D

D is correct. One of the duties of the RMU is to assess the quality of models used to measure portfolio risk, such as through backtesting and validation.

A, B, and C are incorrect. They are the duties of investment managers or the portfolio management function.

Q-63. Solution: B

When comparing the performances of portfolios that are not fully diversified, the appropriate measure to use is Sharpe or M2; Sharpe to rank the performance of the portfolios, M2 to calculate by how much one portfolio outperforms/underperforms the market or the other portfolio. Treynor ratio is appropriate to use when we are comparing many portfolios to form an overall portfolio. Because the number of portfolios combined is high, nonsystematic risk is largely diversified away and beta (rather than standard deviation) can be used to measure risk. This is not the case here. The investor in the question is interested in total risk.

	Manager 1	Manager 2	Market Index
Sharpe	1.61	1.79	1.90
% of P in adjusted P	0.56	0.71	1.00
% of T-bills in adjusted P	0.44	0.29	0.00
Adjusted P return	0.19	0.21	0.22
M^2	-0.03	-0.01	0.00
Treynor's measure	0.24	0.18	0.19

Q-64. Solution: D

Since small movement of an underlying can be captured by delta, large movement should be captured by both delta and gamma. Therefore, re-hedging after significant moves of the underlying stock price is the essence of gamma trading. Credit risk plays an important role in the risk profile of convertible arbitrage hedge funds. Liquidity considerations are essential. Ignorance of this risk can lead to devastating losses as the 2008 financial crisis showed. Gamma trading means frequent

re-hedging of directional exposure after market moves.

Q-65. Solution: D

This position is hedged against interest rate risk, so B) is wrong. It is also hedged against directional movements in the stock, so C) is wrong. The position is long an option (the option to convert the bond into the stock) and so is long implied volatility, so A) is wrong. Long options positions have positive gamma.

Q-66. Solution: C

Global macro strategies take long and short positions based on expectations regarding fundamental changes in global capital markets. The manager in this scenario is engaging in a carry trade by taking a long position in a high-yielding currency (euros) and a short position in a low-yielding currency (Swiss Francs). The manager also expects a fundamental change in the exchange rate between the currencies. Managed futures strategies have a similar philosophy but use futures rather than the underlying assets to execute the strategy.

Q-67. Solution: D

Equity short-selling funds sell stocks not currently owned by the seller in order to take a directional bet that the stock price will decline. However, traditional people who buy stock strategies believe that future stock prices are rising. These short-selling funds tend to be negatively correlated with traditional long-only equity portfolios.

Q-68. Solution: C

The publication lists existing funds, so it must be subject to survivorship bias, because dead funds are not considered. In addition, there is selection bias because the publication focuses on just the popular funds, which are large and likely to have done well. A and B are incomplete. D is also incomplete.

Q-69. Solution: A

The sample is too small: survivorship bias implies that certain non-performing funds (e.g., funds that went out of business) are not included in the sample, it is an issue of the sample.

D is closely related and D is probably true, but D relates to interpretation and A is more directly the implied methodological flaw. So, D is fine, but A is a little better.

Q-70. Solution: B

Statements II, III, and IV are true. Statement I is false — the opposite is true.

Q-71. Solution: C

C is correct. It is the practice of shorting and the leveraging of investors' capital that distinguish hedge funds from conventional long-bias funds.

A is incorrect. An indexed fund is managed to have holdings that mimic a benchmark index and has well-defined performance targets. A hedge fund, on the other hand, can hold substantial short positions in a range of asset categories and apply complex strategies. Therefore, hedge fund portfolio returns are highly dependent on the relevant managers' market timing skills.

B is incorrect. The opacity of hedge fund vehicles persisted for over half a century until the arrival of institutional investors in the new millennium. It was wealthy individual investors of early hedge fund vehicles who demanded their investment in the vehicle be kept private.

D is incorrect. Mutual funds and ETFs must disclose their investment strategies. Hedge funds generally follow proprietary strategies that they see as fundamental to their competitiveness and/or value proposition. They give prospective clients some information to explain their value proposition, but do not disclose everything. Furthermore, they are not obligated to stick to one strategy.

Q-72. Solution: D

D is correct. Hedge fund index returns continued to be less volatile than the S&P Index.

A is incorrect. Institutional investors' expectations have been different than private wealthy individuals. Factors such as risk management, investment process, and operational governance, that contribute to the key consideration of performance persistency, have become more important with the arrival of institutional investors. Institutional investors also focus on how to allocate funds to different hedge fund strategies and need to know if different strategies deliver correlated returns.

B is incorrect. The concentration of assets under management in the industry persisted. Actually, the size difference between the largest funds and the median fund or the smallest funds continued to grow.

C is incorrect. Despite the additional layer of fees charged by funds of hedge funds, the cumulative net (of all fees and expenses) performance of hedge funds did much better than the S&P index.

Q-73. Solution: B

The principal-agent risk arises due to the incentive fee structure and the high water mark (HWM) but can be mitigated

Q-74. Solution: D

D is correct. Because private equity funds trade infrequently, their risk measures—such as volatilities, correlations, and betas—can be too low when computed using reported returns.

A is incorrect. Correlations regarding PEF1 are not reliable (See explanation for D). The correlation between the two ETFs is more reliable.

B is incorrect. Because the volatility of PEF2 is likely understated, it is not clear that PEF2 truly has the best return to risk profile out of the four investments. Even if it did in fact have the best return/risk profile, it is inappropriate to conclude that the investor should allocate all these funds to that one investment. The investor should consider other factors, such as potential diversification benefits from holding a mixture of the four investments as well as diversification benefits with other investments and asset classes that the investor may own. There is not enough information

given to make this conclusion.

C is incorrect. The number of assets alone cannot be a determinant of the risk level of a portfolio. Also, ETFs often hold a very large number of assets, while PEFs can be more concentrated.

Q-75. Solution: C

Generally speaking, with a leveraged fund, an investor will need to evaluate historical and current changes in leverage, as well as the level of liquidity of the portfolio, particularly during times of market stress. Certain strategies may in fact expose an investor to tail risk, so while an investor should inquire whether the manager believes that tail risk exists, and whether or not it is hedged, it is then up to the investor to decide whether to accept the risk unhedged or hedge it on their own. Many funds employ independent risk service providers to report risks to investors, but these firms do not get involved in risk related decision making. And finally, while it is important to know what percentage of the assets is exchange-traded and marked to market, what might be acceptable may differ depending on the strategy of the fund.

Q-76. Solution: C

Investors should assess potential managers and their investment strategies with an objective and unbiased mind. They should not be unduly concerned with a manager's past successes given that past performance is not always indicative of future performance. Risk controls, the business model, and the investment process are all fundamental parts of the due diligence process.

Q-77. Solution: A

A is correct. Investors should make sure they have access to the people at the top of the firm; the actual risk takers and decision makers, so that they have a better sense of what is really going on at that firm. Direct access to founders or senior management is preferred as part of continuing due diligence but if they are not available then the fund should strive to communicate with managers who perform day-to-day investment tasks at the fund.

Communication with investor relations is not sufficient.

B is incorrect. Many funds employ independent risk service providers to report risks to investors, but these firms do not get involved in risk related decision making.

C is incorrect. Investors should evaluate the considered fund's current and historical leverage figures but also understand how and why these figures might deviate from the fund's peers.

D is incorrect. While it is important to know what percentage of the assets is exchange-traded and marked to market, what might be acceptable may differ depending on the strategy of the fund.

Q-78. Solution: A

The fund's Risk Manager reports to the Portfolio Managers with dotted-line reporting to the Traders. In regard to (B), (C) and (D), none is problematic. There is no expectation the service provider is co-located, rather the emphasis is on independence. The risk process is often necessarily complex. And we expect different strategies to incur different styles.

Q-79. Solution: C

This might be a business risk indicator but not a fraud risk indicator. In regard to (A), (B) and (D), each is listed as a potential indicator of fraud that investors should investigate before investing in a fund.

Q-80. Solution: A

"Checklist" is the keyword here. Key themes of the reading include that (i) due diligence is an art and science including both quantitative and qualitative analyses (ii) that should be flexible to the situation and the firm's unique investment strategy but also (iii) that should definitely be comprehensive and holistic where a checklists are a key tool.

The primary mechanism for achieving a comprehensive analysis is the checklist.

Q-81. Solution: B

B is correct. There's no evidence that supports that investors are compensated for fraud risk through higher returns or lower fees.

A is incorrect. Disclosures regarding past regulatory or legal violations are effective in predicting fraud.

C is incorrect. Although certain characteristics, such as conflicts of interest, can predict fraud, we cannot infer that conflicts of interest cause fraud, or that their prohibition would deter fraud.

D is incorrect. In the past, SEC didn't provide access to historical data (contemporaneous data has always been available) but this is not the case anymore. Up until 2007, the investing public's ability to develop and use predictive models based on Form ADV data was potentially limited because the SEC did not provide access to historical data. As a result, the realized benefits of disclosure during that time frame may have been lower. The results of the study given in the reading suggest that improving public access to comprehensive historical disclosures could increase the benefits these disclosures were meant to provide.